



Republic of Rwanda
Ministry of Education



PYTHON PROGRAMMING

GENPP501

Apply python programming fundamentals

Competence

RQF Level: 5

Learning Hours



Credits: 8

Sector: ICT AND MULTIMEDIA

Trade: SOFTWARE DEVELOPMENT

Module Type: General

Curriculum: ICTSDW5001 TVET CERTIFICATE V IN SOFTWARE DEVELOPMENT

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Issue Date: February 2024

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Purpose statement	This specific module describes the skills, knowledge and attitude required to Apply Python Programming. This module is intended to prepare learners pursuing TVET Level 5 in software development. Upon completion of this module, the learner will be able to Prepare python environment, write basic python program and Apply object-driven in python.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	Group project and presentation Individual project /Work
	● Group project and presentation	30%				
	● Individual project /Work	40%				
			Summative Assessment		50%	

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Prepare python environment	1.1 Python tools are properly selected in accordance with computer operating system

	1.2 Python tools are properly installed in accordance with Operating System	
	1.3 Python installation is successfully tested based on output of python version command	
2. Write basic python program	2.1 Python concepts are effectively applied based on python standards	
	2.2 Control structures are properly applied in accordance with python standards	
	2.3 Functions are properly applied in accordance with python standards	
	2.4 Collections are properly applied in accordance with python standards	
	2.5 File handling is properly performed in accordance with python standards	
3. Apply object-driven in python	3.1 OOP Concepts are properly applied in line with python standards	
	3.2 Python date and time concepts are applied according to python standards	
	3.3 Libraries are properly used in accordance with python standards	
	3.4 System automation is properly applied based on specific task	
Knowledge, Skills, and Attitude		
Knowledge	Skills	Attitude

• Description of Algorithm and data structure • Description of Python libraries • Description of Python language features • Description of Python syntax and semantics • Description of concepts Python programming	• Applying python tools • Installation of python software tools • Configuration of python virtual environment • Running python version command • Checking python interpreter • Testing package manager • Using operators • Applying python practices • Applying Conditional Statements • Applying looping Statements • Creation of function • Performing write/create /read and delete file • Applying OOP Concepts • Performing relative timedeltas • Selecting Python automation library • Integration of script with Deployment Process	• Being flexible • Team spirit • Paying Attention to details • Being responsible • Proficiency about Programming skills • Having critical thinking • Being Problem solver • Being self-motivated • Trustworthy to details
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Learning outcomes	At the end of the module the learner will be able to: 1. Prepare python environment 2. Write basic python program 3. Apply object-driven in python
Learning outcome 1: Prepare python environment	Learning hours: 15
Indicative content	
● Selection of Python tools ✓ Python programming overview  Definition  Benefits  Characteristics of python ✓ Applications of python  Data science  Software development  Automation  Data analytics ✓ Identifications of python tools ● Installation of Python tools ✓ Identification of computer system requirements  Hardware requirement  Software requirement ✓ Install python software tools ✓ Configure python virtual environment	

- **Testing python installation**
 - ✓ Run python version command
 - ✓ Check python interpreter
 - ✓ Test package manager

Resources required for the learning outcome	
Equipment	● Computer, Projector
Materials	● Internet
Tools	● Text editor, Python(latest and stable version), IDE
Facilitation techniques	● Group discussion ● Brainstorming ● Practical exercises in group work ● Individual Practical exercises
Formative assessment methods	● Written assessment ● Oral questions ● Performance assessment

Learning outcome 2: Write basic python program

Learning hours: 45

Indicative content

- **Applying python basic concepts**

- ✓ Data Types
- ✓ Variables
- ✓ Comments
- ✓ Operators

- **Applying python control structures**

- ✓ Conditional Statements
- ✓ Looping Statements
- ✓ Use of Jump Statements
 - ✚ Break
 - ✚ Continue
 - ✚ Pass

- **Application of functions in Python**

- ✓ Definition of Function
- ✓ Types of functions
 - ✚ built-in function
 - ✚ user defined function
- ✓ Create a function
 - ✚ Arguments
 - ✚ Default parameter value

- ✚ Passing a list as an argument

- ✚ Calling a function

- ✓ Apply special purpose

functions

- ✚ Lambda

- ✚ Apply Python Generators

- ✚ Apply Python Closures

- ✚ Apply Python Decorators

- ✚ Recursive function

- ✚ Higher-order function

- **Application of Python Collections**

- ✓ Description of collection types

- ✚ Lists

- ✚ Tuples

- ✚ Dictionaries

- ✚ Sets

- ✚ Frozen Set

- ✚ ChainMaps

- ✚ Deques

- ✓ Specialized tools from the

collections module

- ✚ Counter

- ✚ OrderedDict

- ✚ defaultdict

- ✓ Perform common operations

- ✚ Adding and Removing
Elements

- ✚ Accessing and Iterating
Over Elements

- ✚ Filtering and Sorting

- ✚ Set Operations and
Counting

- ✚ Stack and Queue
Operations

- **Perform File handling**

- ✓ File Handling libraries

- ✚ os

- ✚ pathlib

- ✚ shutil

- ✚ pandas

- ✓ Practice to read file

- ✚ Open file

- ✚ Read file permission

- ✓ Perform write/create file

- ✚ Create a new file

- ✚ Write to existing file

- ✓ Perform delete file

- ✚ Remove file

- ✚ Delete folder

- ✓ Apply python best practices

- ✚ Readability and Style

- ✚ Use of Built-in Features

- ✚ Efficiency and Memory

- Usage

 Error Handling and
Testing

Resources required for the indicative content

Equipment	• Computer, Projector
Materials	• Internet
Tools	• Text editor, Python (latest and stable version),IDE
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral questions • Performance assessment

Learning outcome 3: Apply object-driven in python

Learning hours: 40

Indicative content

- **Application of OOP Concepts**

- ✓ Object
- ✓ Python class
- ✓ Inheritance
- ✓ Polymorphism
- ✓ Encapsulation

- **Application of python Date and time concepts**

- ✓ Description of date and time
 -  Datetime
 -  Dateutil
 -  Arrow
 -  Pendulum
 -  Python-tzdata

- ✓ Set Time zones
- ✓ Formatting and parsing
- ✓ Perform relative timedeltas

- **Application of Python Libraries**

- ✓ Description of Python Library
- ✓ Python standard library
 -  Matplotlib
 -  Numpy
 -  Pandas

- ✓ Use of libraries
 - + Importing Libraries
 - + Accessing Functionality
 - + Understanding Scope according to the name space

- **Automation of application post deployment tasks**

- ✓ Identification of tasks to automate
 - + Database migrations
 - + Configuration file updates
 - + Service restarts
 - + Testing and verification
 - + Logging and notifications
- ✓ Identification of tasks to be prioritized
 - + Repetitive
 - + Time-consuming
 - + Error-prone
 - + Critical for deployment speed
- ✓ Selection of a Python Automation Library
 - + Fabric
 - + Ansible
 - + SaltStack
 - + Boto3
 - + VSphere Automation SDK for Python for VMware vSphere environments.
 - + Consider factors
- ✓ Develop the Python Script
 - + Use library functions
 - + Structure logically
 - + Logging and output
- ✓ Integrate script with Deployment Process
 - + Trigger method to initiate the Python script post-deployment

- ✚ Direct execution after deployment completion
 - ✚ Integration with CI/CD pipelines
 - ✚ Scheduled execution at specific intervals
 - ✚ Implement security measures to restrict script access and control sensitive information.
- ✓ Testing and Monitoring
- ✚ Thorough testing
 - ✚ Monitor script logs execution
 - ✚ Refine and improve

Resources required for the indicative content

Equipment	• Computer, Projector
Materials	• internet
Tools	• Text editor , Python(latest and stable version), IDE , Jupyter notebook
Facilitation techniques	• Group discussion • Brainstorming • Practical exercises in group work • Individual Practical exercises
Formative assessment methods	• Written assessment • Oral question • Performance assessment

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